

TOBY S. COX

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EDUCATION

CLEMSON UNIVERSITY

Clemson, SC

Bachelor of Science in Computer Science, Minor in Artificial Intelligence

May 2027

Bachelor of Science in Mathematical Sciences, Minor in Physics

GPA: 4.00/4.00

RELEVANT COURSEWORK

Machine Learning: Implementation and Evaluation (A) Theory of Probability (A) Applied Data Science (A)
Introduction to Software Engineering (A) Data Structures and Algorithms (A) Calculus of Several Variables (A)
Linear Programming (A) Artificial Intelligence (A) Introduction to Ordinary Differential Equations (A)

EXPERIENCE

Undergraduate Teaching Assistant

Clemson University

CPSC 1020 (Computer Science II)

Fall 2024

- Led twice-weekly lab sections with thirty students, covering topics such as Linux, C++ syntax, data structures and algorithms, file input/output, classes, and inheritance
- Held office hours to provide individualized assistance to students
- Graded student submissions for a class of over one hundred students

Undergraduate Research Assistant

Clemson University

SCALab (Advisors: Dr. Rong Ge, Nathan Jones)

Spring 2025-Present

- Trained specialized diffusion U-Net models using GPUs and multi-CPU programming on an HPC cluster
- Achieved upwards of 8x improvement in circuit fidelity over current methods
- Poster accepted to ACM/IEEE SC 2025 conference

SELECTED PROJECTS

FABRIC Operations- Accelerating Innovation and Research (*Undergraduate Researcher, REU*) Summer 2025

- Leveraged the FABRIC testbed for national, high-scale scientific computing
- Analyzed data transfer rates using iPerf3 across different congestion control protocols

HackMIT (*Team Lead, Hackathon*)

Fall 2025

- Leveraged YOLO and MediaPipe to create a real-time synthesizer from paper
- Awarded 1st place in Suno's "Best Musical Hack" category, out of ~300 total teams at the event

HoyaHacks (*Team Lead, Hackathon*)

Spring 2026

- Created monitoring software for HPC clusters that tracks most accessed files, with interactive dashboard
- Awarded "Best Use of MongoDB" out of ~120 total teams at the event

MIT iQuHack (*Participant, Hackathon*)

Spring 2026

- Engineered diverse features and implemented ensemble deep learning for quantum simulation predictions
- Awarded 2nd place in Quantum Rings' "Circuit Fingerprint Challenge" out of a global field of top students

NASA SpaceApps (*Team Lead, Hackathon*)

Fall 2025

- Modeled differential equations via approximation and iteration to model asteroid impacts and their effects
- Awarded 1st place in the 3rd biggest local event in the US for "Most Inspirational Hack"

Hands-On Quantum Computing (*Undergraduate Researcher, Creative Inquiry*)

Spring 2025

- Conducted formal verification of quantum algorithms using Isabelle/HOL
- Implemented quantum algorithms in qBraid, using Qiskit library for Python

SKILLS

PROGRAMMING LANGUAGES

- Python, C++, JavaScript, C, Java, R, Julia, Isabelle/HOL, HTML, CSS, TypeScript, Sage, SQL, Prolog

SOFTWARE/PACKAGES

- Google Suite, Microsoft Office, AWS, Postman, Qiskit, TensorFlow, PyTorch, NodeJS, MongoDB

MISCELLANEOUS

- German, LaTeX/Overleaf, Unix/Linux, qBraid

HONORS AND AWARDS

Phi Kappa Phi

President's List (Clemson) (Fall 2023 – Present)

Upsilon Pi Epsilon

Clemson University Honors College (Fall 2023 – Present)